TRAIN RESERVATION SYSTEM — COMPLETE SQL SOURCE CODE

Student: Shivansh (UID: 25BCA10433) | **Section:** 25BCA 6B | **Engine:** MySQL 8.0 Enterprise

1. DATABASE CREATION & CORE DDL SCHEMA

```
-- ========================================================================




 train_reservation_db;
 train_reservation_db;


 Users (
    user_id ,
    full_name ,
    email ,
    phone_number ,
    password_hash ,
    user_role ,
    created_at
);


 Trains (
    train_id ,
    train_number ,
    train_name ,
    train_type ,
    total_coaches
);


 Stations (
    station_id ,
    station_code ,
    station_name ,
    city ,
    state ,
    zone
);


 Train_Routes (
    route_id ,
    train_id ,
    station_id ,
    stop_sequence ,
    distance_from_source_km ,
    halt_duration_minutes ,
     (train_id)  Trains(train_id) ,
     (station_id)  Stations(station_id)
);


 Schedules (
    schedule_id ,
    train_id ,
    departure_time ,
    arrival_time ,
    runs_on_days ,
     (train_id)  Trains(train_id)
);
-- DATABASE: TRAIN RESERVATION AND TICKET BOOKING MANAGEMENT SYSTEM-- TARGET ENGINE: MySQL 8.0+
InnoDB (3NF Normalized)--
========================================================================CREATE DATABASE IF NOT
EXISTSUSE-- 1. USERS TABLECREATE TABLEINT AUTO_INCREMENT PRIMARY KEYVARCHAR(100) NOT NULL
```

```
VARCHAR(100) UNIQUE NOT NULLVARCHAR(15) UNIQUE NOT NULLVARCHAR(255) NOT NULLENUM('Passenger',
'Admin') DEFAULT 'Passenger'TIMESTAMP DEFAULT CURRENT_TIMESTAMP-- 2. TRAINS TABLECREATE TABLEINT
AUTO_INCREMENT PRIMARY KEYVARCHAR(10) UNIQUE NOT NULLVARCHAR(100) NOT NULLENUM('Superfast',
'Express', 'Rajdhani') NOT NULLINT NOT NULL DEFAULT 10-- 3. STATIONS TABLECREATE TABLEINT
AUTO_INCREMENT PRIMARY KEYVARCHAR(10) UNIQUE NOT NULLVARCHAR(100) NOT NULLVARCHAR(50) NOT
NULLVARCHAR(50) NOT NULLVARCHAR(10) NOT NULL-- 4. TRAIN_ROUTES TABLECREATE TABLEINT AUTO_INCREMENT
PRIMARY KEYINT NOT NULLINT NOT NULLINT NOT NULLDECIMAL(7,2) NOT NULLINT DEFAULT 2FOREIGN
KEYREFERENCESON DELETE CASCADEFOREIGN KEYREFERENCESON DELETE RESTRICT-- 5. SCHEDULES TABLECREATE
TABLEINT AUTO_INCREMENT PRIMARY KEYINT NOT NULLTIME NOT NULLTIME NOT NULLVARCHAR(50) NOT
NULLFOREIGN KEYREFERENCESON DELETE CASCADE
```

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2. RESERVATION, COACHES, SEATS & TRANSACTION DDL

```
-- 6. COACHES TABLE
 Coaches (
    coach_id ,
    train_id ,
    coach_code ,
    coach_class ,
    total_capacity ,
     (train_id)  Trains(train_id)
);


 Seats (
    seat_id ,
    coach_id ,
    seat_number ,
    berth_type ,
     (coach_id)  Coaches(coach_id)
);


 Passengers (
    passenger_id ,
    full_name ,
    age ,
    gender ,
    id_proof_type ,
    id_proof_number
);


 Bookings (
    booking_id ,
    pnr_number ,
    user_id ,
    schedule_id ,
    source_station_id ,
    destination_station_id ,
    total_fare ,
    booking_status ,
    booking_date ,
     (user_id)  Users(user_id),
     (schedule_id)  Schedules(schedule_id),
     (source_station_id)  Stations(station_id),
     (destination_station_id)  Stations(station_id)
);


 Tickets (
    ticket_id ,
    booking_id ,
    passenger_id ,
    seat_id ,
    travel_date ,
    ticket_status ,
     (booking_id)  Bookings(booking_id) ,
     (passenger_id)  Passengers(passenger_id),
     (seat_id)  Seats(seat_id)
);


 Payments (
    payment_id ,
    booking_id ,
    transaction_ref ,
```

```
CREATE TABLEINT AUTO_INCREMENT PRIMARY KEYINT NOT NULLVARCHAR(10) NOT NULLENUM('SL', '3A', '2A',
'1A', 'CC') NOT NULLINT NOT NULL DEFAULT 72FOREIGN KEYREFERENCESON DELETE CASCADE-- 7. SEATS
TABLECREATE TABLEINT AUTO_INCREMENT PRIMARY KEYINT NOT NULLINT NOT NULLENUM('Lower', 'Middle',
'Upper', 'Side Lower', 'Side Upper') NOT NULLFOREIGN KEYREFERENCESON DELETE CASCADE-- 8. PASSENGERS
TABLECREATE TABLEINT AUTO_INCREMENT PRIMARY KEYVARCHAR(100) NOT NULLINT NOT NULLENUM('Male',
'Female', 'Other') NOT NULLVARCHAR(30) NOT NULLVARCHAR(50) NOT NULL-- 9. BOOKINGS TABLECREATE
TABLEINT AUTO_INCREMENT PRIMARY KEYVARCHAR(10) UNIQUE NOT NULLINT NOT NULLINT NOT NULLINT NOT
NULLINT NOT NULLDECIMAL(10,2) NOT NULLENUM('Confirmed', 'Pending', 'Cancelled') DEFAULT
'Confirmed'TIMESTAMP DEFAULT CURRENT_TIMESTAMPFOREIGN KEYREFERENCESFOREIGN KEYREFERENCESFOREIGN
KEYREFERENCESFOREIGN KEYREFERENCES-- 10. TICKETS TABLECREATE TABLEINT AUTO_INCREMENT PRIMARY KEYINT
NOT NULLINT NOT NULLINT NOT NULLDATE NOT NULLENUM('CNF', 'RAC', 'WL', 'CAN') DEFAULT 'CNF'FOREIGN
KEYREFERENCESON DELETE CASCADEFOREIGN KEYREFERENCESFOREIGN KEYREFERENCES-- 11. PAYMENTS TABLECREATE
TABLEINT AUTO_INCREMENT PRIMARY KEYINT NOT NULLVARCHAR(50) UNIQUE NOT NULLENUM('UPI', 'Credit
Card', 'Net Banking', 'Wallet') NOT NULLDECIMAL(10,2) NOT NULLENUM('Success', 'Failed', 'Refunded')
DEFAULT 'Success'TIMESTAMP DEFAULT CURRENT_TIMESTAMPFOREIGN KEYREFERENCESON DELETE CASCADE-- 12.
CANCELLATIONS TABLECREATE TABLEINT AUTO_INCREMENT PRIMARY KEYINT NOT NULLTIMESTAMP DEFAULT
CURRENT_TIMESTAMPDECIMAL(10,2) NOT NULLENUM('Processed', 'Pending', 'Failed') DEFAULT
'Processed'FOREIGN KEYREFERENCESON DELETE CASCADE
```

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3. STORED PROCEDURES, TRIGGERS & ANALYTICAL VIEWS

```
-- --------------------------------------------------------------------


 //
 sp_BookTrainTicket(
     p_user_id ,
     p_schedule_id ,
     p_source_id ,
     p_dest_id ,
     p_fare ,
     p_pass_name ,
     p_pass_age ,
     p_pass_gender ,
     p_seat_id ,
     p_travel_date ,
     p_generated_pnr
)

     v_booking_id ;
     v_passenger_id ;

    ;


     p_generated_pnr = ((() * 10000000000), 10, );


     Bookings (pnr_number, user_id, schedule_id, source_station_id, destination_station_id,
total_fare, booking_status)
     (p_generated_pnr, p_user_id, p_schedule_id, p_source_id, p_dest_id, p_fare, );

     v_booking_id = ();


     Passengers (full_name, age, gender, id_proof_type, id_proof_number)
     (p_pass_name, p_pass_age, p_pass_gender, , (, (()*900000)));

     v_passenger_id = ();


     Tickets (booking_id, passenger_id, seat_id, travel_date, ticket_status)
     (v_booking_id, v_passenger_id, p_seat_id, p_travel_date, );

    ;
 //
 ;




 //
 trg_AfterBookingCancelled
 Bookings


     NEW.booking_status =   OLD.booking_status !=
         Cancellations (booking_id, cancellation_date, refund_amount, refund_status)
         (NEW.booking_id, (), NEW.total_fare * 0.80, );

         Tickets  ticket_status =   booking_id = NEW.booking_id;
    ;
 //
 ;
```

```
-- STORED PROCEDURE: ATOMIC TICKET RESERVATION TRANSACTION-- ------------------------------------
-----------------------------DELIMITERCREATE
PROCEDUREININTININTININTININTINDECIMAL(10,2)INVARCHAR(100)ININTINVARCHAR(10)ININTINDATEOUTVARCHAR(1
0)BEGINDECLAREINTDECLAREINTSTART TRANSACTION-- Generate unique 10-digit PNRSETLPADFLOORRAND'8'--
Create Booking RecordINSERT INTOVALUES'Confirmed'SETLAST_INSERT_ID-- Create Passenger RecordINSERT
INTOVALUES'Aadhaar Card'CONCAT'ID-'FLOORRANDSETLAST_INSERT_ID-- Issue TicketINSERT
INTOVALUES'CNF'COMMITENDDELIMITER-- ------------------------------------------------------------------
------- TRIGGER: AUTOMATIC REFUND CREATION ON CANCELLATION-- ------------------------------------
-----------------------------DELIMITERCREATE TRIGGERAFTER UPDATE ONFOR EACH
ROWBEGINIF'Cancelled'AND'Cancelled'THENINSERT INTOVALUESNOW'Processed'UPDATESET'CAN'WHEREEND
IFENDDELIMITER-- ------------------------------------------------------------------------ VIEWS & B-
TREE PERFORMANCE INDEXES-- ------------------------------------------------------------------------
CREATE VIEWASSELECTASASFROMJOINONJOINONJOINONJOINONJOINONJOINONCREATE INDEXONCREATE INDEXON
```